Optimal Trading Strategies for Energy Hubs Equipped with Power-to-Gas Technology

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***Abstract*—An energy hub, which is considered as a unit where multiple energy carriers can be converted, conditioned, and stored, becomes a key element in future energy networks due to the emergence of the integrated energy system. Much is unknown about the management of energy hubs, for instance, how much energy to convert, how much energy to store, and how relevant it is to different factors such as prices. We answer these questions by formulating the trading decision making problem for energy hubs as a dynamic program. For an extreme setting that incorporates uncertainty, we prove the optimality of a two-threshold policy of the energy conversion. Under the assumption with parameters that leads to the normal case, we also establish two-threshold structures of the optimal operation of the energy conversion and the energy storage.**

***Index Terms*—Dynamic programming, energy hub, power-to-gas, uncertainty.**

I. Introduction

A key element in the vision of future energy networks is the so-called energy hub. An energy hub is considered as a unit where multiple energy carriers can be converted, conditioned, and stored. It represents an interface between different energy infrastructures and loads. The flexible combination of different energy carriers using the conversion and storage technology offers a powerful approach for various system improvements. When designing multi-energy systems, it is useful to investigate energy hubs among various energy carriers [1]. With the maturity of power-to-gas (P2G) technology, the installment of P2G in energy hubs becomes practical. P2G provides not only a tool to convert electricity to natural gas but also a promising solution for the large-scale and long-term storage of electricity. Energy hubs act as retailers of different energy carriers due to the emergence of the integrated energy system. Optimizing the trading decisions for energy hubs aims to obtain the optimal involvement in electricity and other energy markets, and to maximize the expected profit. A few studies have discussed the operation of energy hubs. A bilevel stochastic programming problem model of the decision making of an energy hub manager is presented in [2]. Reference [3] discusses the planning of energy hubs systems considering the uncertainty. From this perspective, P2G has been rarely considered in energy hubs. On the other hand, studies that have discussed the use of P2G, e.g., [4], [5], focus on a system perspective.

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On the above premises, we construct a dynamic programming framework to study the trading decision making problem faced by the manager of energy hubs equipped with P2G who seeks to optimally operate the energy hubs, and trade electricity as well as natural gas in the wholesale markets. There have been widespread studies on the application of dynamic programming in the decision making problems of power systems. A few recent works take a dynamic programming approach to derive the optimal operation of storage [6], [7]. An optimization problem of hydro system solved by dynamic programming is studied in [8]. Moreover, reference [9] discusses the optimal operation of a virtual power plant considering lagged commitments in a dynamic programming framework.

Day-ahead prices are realized simultaneously. We can view day-ahead price vectors as realizations of intraday randomness, which allows us to describe the dynamics of the hourly electricity prices by a discrete state transition [8]. Our structural analysis helps the managers of energy hubs to optimize their trading decisions, and assists them in optimally determining the energy conversion between electricity and natural gas as well as the operation of the storage.

The rest of the paper is organized as follows. In Section II, we formulate the problem in a dynamic programming framework. In Section III, we solve the problem analytically, and introduce some threshold policies for the energy conversion and the energy storage. We present numerical results of the thresholds in Section IV. Section V concludes the paper.

II. Problem Formulation

We study the trading strategy for energy hubs. The energy hubs can discharge their storage or charge their storage. As for the energy conversion, the energy hubs have the options of converting electricity to natural gas, i.e., in P2G mode, and converting natural gas to electricity, i.e., in G2P mode. Fig. 1 depicts the coupling of those two energy carriers. We assume that the energy hub is small enough compared to the market, so that the decisions would not affect the prices. The manager makes trading decisions over a finite horizon $t \in \mathbb{T} := \{0, 1, \ldots, T\}$, and any energy left in the storage at the end of the scheduling horizon is worthless. Moreover, we assume that the capacity of the storage and the capacity of the

conversion are limited, but the line capacities of both energy carrier are large enough.

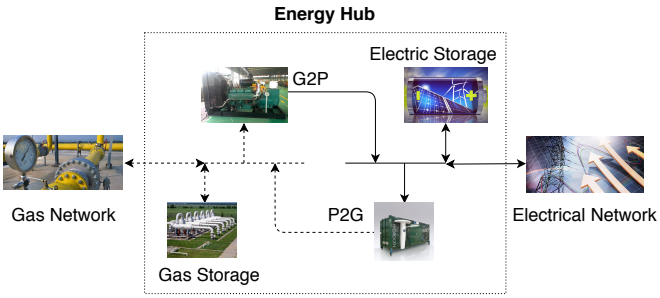


Fig. 1: General scheme of an energy hub.

A. Modeling the Uncertainty

We consider the prices of gas and electricity as random processes. Wind generation depending on wind speed is also random. We model the prices as a mean-reverting process. Accordingly, we assume a discrete time version as the following equations:

$$P^{\mathrm{G}}_{t+1} = P^{\mathrm{G}}_{t} + k^{\mathrm{G}}(\mu^{\mathrm{G}} - P^{\mathrm{G}}_{t}) + \hat{P}^{\mathrm{G}}_{t+1}, \tag{1}$$

$$P^{\mathrm{E}}_{t+1} = P^{\mathrm{E}}_{t} + k^{\mathrm{E}}(\mu^{\mathrm{E}} - P^{\mathrm{E}}_{t}) + \hat{P}^{\mathrm{E}}_{t+1}, \tag{2}$$

where k^{G} and k^{E} are the mean-reversion rates which govern the speed of reversion to the mean price μ^{G} and μ^{E} respectively; The random variables $\hat{P}^{\mathrm{G}}_{t+1}$ and $\hat{P}^{\mathrm{E}}_{t+1}$ are i.i.d with distribution $\mathcal{N}(0, \sigma^2_{\mathrm{G}})$ and $\mathcal{N}(0, \sigma^2_{\mathrm{E}})$, which include noise in the evolution of price. (1) and (2) can be converted to AR(1) process with $\alpha_1 = 1 - k$ and $\hat{P}_{t+1} \sim \mathcal{N}(\mu k, \sigma^2)$.

Unlike the price model, a seasonality term is used to correct wind generation. We model the wind generation as the sum of an AR(1) process and a seasonality function $S(t)$:

$$W_{t+1} = \alpha_{\mathrm{wind}} W_t + \hat{W}_{t+1} + S(t), \tag{3}$$

where α_{wind} is a scalar, and $\hat{W}_{t+1} \sim \mathcal{N}(0, \sigma^2_{\mathrm{wind}})$ is an i.i.d term. The seasonality $S(t)$ is

$$S(t) = A + \gamma cos[(t + \omega)2\pi/24], \tag{4}$$

where A is a constant level, γ and ω are the magnitude and the phase shift of the hourly seasonality.

B. State Variables

State variables with subscript t are known in period t, but unknown in the earlier period. State S_t can be defined as $(W_t, P^{\mathrm{G}}_t, P^{\mathrm{E}}_t, IG_t, IE_t)$, which includes:

- W_t : the wind generation produced in period t in MWh;
- P^{G}_t : the price of gas in period t in \$/mcf;
- P^{E}_t : the price of electricity in period t in \$/MWh;
- IG_t : the state of charge of the gas storage in period t in mcf.
- IE_t : the state of charge of the electric storage in period t in MWh.

C. Decision Variables

- E_t : the quantity of electricity to sell or buy in period t in MWh.
- G_t : the quantity of gas to sell or buy in period t in mcf.
- A^{p2g}_t : the operation of P2G or G2P in period t in MWh. Although G2P and P2G are not the same equipment, they will never both be on at the same time. In order to reduce the decision space, we use one decision variable to denote their operation. If $A^{\mathrm{p2g}}_t < 0$, the converter is operating in G2P mode; if $A^{\mathrm{p2g}}_t \geq 0$, the converter is operating in P2G mode.

It should be noted that two equations of energy balance should be considered, which give the operation of storage:

$$A^{\mathrm{e}}_t = W_t - A^{\mathrm{p2g}}_t - E_t, \tag{5}$$

$$A^{\mathrm{g}}_t = F^{\mathrm{p2g}}_t - G_t, \tag{6}$$

where A^{e}_t and A^{g}_t are the charging and discharging rates of storage in MWh and mcf respectively; F^{p2g}_t is the gas flow injection associated with the conversion of A^{p2g}_t. A control sequence $\{\pi_t(E_t, G_t, A^{\mathrm{p2g}}_t), t \in \mathbb{T}\}$ is used to denote decisions.

D. Dynamic Equation

The wind generation and prices are determined by the random process. The state of charge of gas and electric storage is determined by the charging or discharging rates where if $A^{\mathrm{e}}_t < 0$ or $A^{\mathrm{g}}_t < 0$, the storage is discharging. The transitions of storage are given below:

$$LE_{t+1}(S_t, \pi_t) = \begin{cases} \min\{IE_{\max}, IE_t + \eta_{\mathrm{e}} A^{\mathrm{e}}_t\} & \text{if, } A^{\mathrm{e}}_t \geq 0, \\ \max\{0, IE_t + \frac{1}{\eta_{\mathrm{e}}} A^{\mathrm{e}}_t\} & \text{if, } A^{\mathrm{e}}_t < 0, \end{cases} \tag{7}$$

$$LG_{t+1}(S_t, \pi_t) = \begin{cases} \min\{IG_{\max}, IG_t + \eta_{\mathrm{g}} A^{\mathrm{g}}_t\} & \text{if, } A^{\mathrm{g}}_t \geq 0, \\ \max\{0, IG_t + \frac{1}{\eta_{\mathrm{g}}} A^{\mathrm{g}}_t\} & \text{if, } A^{\mathrm{g}}_t < 0, \end{cases} \tag{8}$$

where $IE_{\max}$ and $IG_{\max}$ are the maximum capacity of electric and gas storage; η_{e} and η_{g} are the efficiency of electric and gas storage. The state dynamic equation is given by:

$$\begin{aligned} S_{t+1} &= T(S_t, \pi_t) \\ &= (W_{t+1}(W_t), P^{\mathrm{G}}_{t+1}(P^{\mathrm{G}}_t), \\ &\quad P^{\mathrm{E}}_{t+1}(P^{\mathrm{E}}_t), LG_{t+1}(IG_t, \pi_t), LE_{t+1}(IE_t, \pi_t)). \end{aligned} \tag{9}$$

E. Revenue to Go

In period t, the intermediate revenue is given by:

$$R(S_t, \pi_t) = E_t P^{\mathrm{E}}_t + G_t P^{\mathrm{G}}_t. \tag{10}$$

The decision variables are feasible if and only if all the following constraints hold:

$$-A^{\max}_{\mathrm{g2p}} \leq A^{\mathrm{p2g}}_t \leq A^{\max}_{\mathrm{p2g}}, \tag{11a}$$

$$-A^{\max}_{\mathrm{e}} \leq A^{\mathrm{e}}_t \leq A^{\max}_{\mathrm{e}}, \tag{11b}$$

$$-A^{\max}_{\mathrm{g}} \leq A^{\mathrm{g}}_t \leq A^{\max}_{\mathrm{g}}, \tag{11c}$$

$$F^{\mathrm{p2g}}_t = 3.412 \eta_{\mathrm{p2g}} A^{\mathrm{p2g}}_t, \text{ if, } A^{\mathrm{p2g}}_t \geq 0, \tag{11d}$$

$$F^{\mathrm{p2g}}_t = \frac{3.412}{\eta_{\mathrm{p2g}}} A^{\mathrm{p2g}}_t, \quad \text{if, } A^{\mathrm{p2g}}_t < 0, \tag{11e}$$

where (11d) and (11e) give the conversion between the quantity of electricity and the quantity of gas.

Our goal is to maximize the total revenue over the time horizon $\mathbb{T}$. A feasible decision set Π, which is determined by (11a), (11b) and (11c), will not change with states or actions. The total revenue can be defined as:

$$\max_{\pi\in\Pi}\sum_{t=0}^{T}\mathbb{E}\left[R(S_t,\pi_t)|S_0\right], \tag{12}$$

where S_0 is the initial state.

A recursive function can be formulated to solve the problem in a dynamic programming framework. The maximum revenue to go in period t given S_t can be obtained as:

$$J_t(S_t)=\max_{\pi_t\in\Pi}\left\{R(S_t,\pi_t)+\mathbb{E}\left[J_{t+1}(S_{t+1})|(S_t,\pi_t)\right]\right\}, \tag{13}$$

where the boundary condition is: $J_{T+1}(S_{T+1})=0,\forall S_{T+1}$.

III. Optimal Policies

In this section, we solve the problem using dynamic programming. We assume the mean-revision rate of price model is less than 1. To ensure boundedness, we consider 95% of the values of the normal distribution that lies within two standard deviations, and scale the probability density by $\frac{1}{0.95}$. For the problem initialization, we assume the initial prices in period $t=0$ are the mean price.

A. The Extreme Case

Lemma 1. *For every period $t\in\mathbb{T}$, P_t^{G} and P_t^{E} are bounded, given μ, σ and k.*

Proof. P_t^{G} is bounded by $\mu^{\mathrm{G}}-2t\sigma_{\mathrm{G}}+2(t-1)k\sigma_{\mathrm{G}}$ and $\mu^{\mathrm{G}}+2t\sigma_{\mathrm{G}}-2(t-1)k\sigma_{\mathrm{G}}$. In period $t=1$, we have P_1^{G} = P_0^{G} + $\hat{P}_1^{\mathrm{G}}$. Using the assumptions mentioned before, it turns to $P_1^{\mathrm{G}}\in$ [μ^{G} − $2\sigma_{\mathrm{G}}$, μ^{G} + $2\sigma_{\mathrm{G}}$]. Hence the hypothesis holds for $t=1$.
Suppose this hypothesis holds for all periods 1,..., $t-1$. We next prove that for period t. $\mu^{\mathrm{G}}-P_{t-1}^{\mathrm{G}}$ is bounded by $2(t-1)\sigma_{\mathrm{G}}-2(t-2)k\sigma_{\mathrm{G}}$ and $-2(t-1)\sigma_{\mathrm{G}}+2(t-2)k\sigma_{\mathrm{G}}$. This leads to:

$$\begin{aligned}
P_t^{\mathrm{G}} &\geq \mu^{\mathrm{G}}-2(t-1)\sigma_{\mathrm{G}}+2(t-2)k\sigma_{\mathrm{G}}\\
&\quad+k[2(t-1)\sigma_{\mathrm{G}}-2(t-2)k\sigma_{\mathrm{G}}]-2\sigma_{\mathrm{G}}\\
&\geq \mu^{\mathrm{G}}-2t\sigma_{\mathrm{G}}+2(t-1)k\sigma_{\mathrm{G}}+2(t-2)(1-k)\sigma_{\mathrm{G}}\\
&\geq \mu^{\mathrm{G}}-2t\sigma_{\mathrm{G}}+2(t-1)k\sigma_{\mathrm{G}},\\
P_t^{\mathrm{G}} &\leq \mu^{\mathrm{G}}+2(t-1)\sigma_{\mathrm{G}}-2(t-2)k\sigma_{\mathrm{G}}\\
&\quad+k[-2(t-1)\sigma_{\mathrm{G}}+2(t-2)k\sigma_{\mathrm{G}}]+2\sigma_{\mathrm{G}}\\
&\leq \mu^{\mathrm{G}}+2t\sigma_{\mathrm{G}}-2(t-1)k\sigma_{\mathrm{G}}-2(t-2)(1-k)\sigma_{\mathrm{G}}\\
&\leq \mu^{\mathrm{G}}+2t\sigma_{\mathrm{G}}-2(t-1)k\sigma_{\mathrm{G}}.
\end{aligned}$$

P_t^{G} is bounded, and it holds for all $t\in\mathbb{T}$. Similarly, P_t^{E} is bounded. □

Lemma 1 implies that the largest variation from mean price of price at period t is bounded by k and σ. The maximum variation from mean price of price happens at period $t=T$. This would lead to two extreme conditions that:

$$\begin{cases} P_T^{\mathrm{G}}=\mu^{\mathrm{G}}-2T\sigma_{\mathrm{G}}+2(T-1)k\sigma_{\mathrm{G}},\\ P_T^{\mathrm{E}}=\mu^{\mathrm{E}}+2T\sigma_{\mathrm{E}}-2(T-1)k\sigma_{\mathrm{E}},\end{cases}$$

and

$$\begin{cases} P_T^{\mathrm{G}}=\mu^{\mathrm{G}}+2T\sigma_{\mathrm{G}}-2(T-1)k\sigma_{\mathrm{G}},\\ P_T^{\mathrm{E}}=\mu^{\mathrm{E}}-2T\sigma_{\mathrm{E}}+2(T-1)k\sigma_{\mathrm{E}}.\end{cases}$$

Theorem 2. *There exists an optimal policy π^* = $(\pi_1^*,\ldots,\pi_T^*)$ characterized by μ^{G}, σ_{G}, μ^{E}, and σ_{E}:*

1) *If $3.412/\eta_{\mathrm{p2g}}\left[\mu^{\mathrm{G}}+2T\sigma_{\mathrm{G}}-2(T-1)k\sigma_{\mathrm{G}}\right]\leq\mu^{\mathrm{E}}-2T\sigma_{\mathrm{E}}+2(T-1)k\sigma_{\mathrm{E}}$, then $A_t^{*\mathrm{p2g}}=-A_{\mathrm{g2p}}^{\max}$, $\forall t\in\mathbb{T}$.*
2) *If $3.412\eta_{\mathrm{p2g}}\left[\mu^{\mathrm{G}}-2T\sigma_{\mathrm{G}}+2(T-1)k\sigma_{\mathrm{G}}\right]\geq\mu^{\mathrm{E}}+2T\sigma_{\mathrm{E}}-2(T-1)k\sigma_{\mathrm{E}}$, then $A_t^{*\mathrm{p2g}}=A_{\mathrm{p2g}}^{\max}$, $\forall t\in\mathbb{T}$.*

Proof. If $3.412/\eta_{\mathrm{p2g}}\left[\mu^{\mathrm{G}}+2T\sigma_{\mathrm{G}}-2(T-1)k\sigma_{\mathrm{G}}\right]\leq\mu^{\mathrm{E}}-2T\sigma_{\mathrm{E}}+2(T-1)k\sigma_{\mathrm{E}}$, consider prices of two energy carriers using the same unit, which leads to $3.412/\eta_{\mathrm{p2g}}\times P_t^{\mathrm{G}}<P_t^{\mathrm{E}},\forall t$. Because $\mu^{\mathrm{G}}+2T\sigma_{\mathrm{G}}-2(T-1)k\sigma_{\mathrm{G}}$ is the upper bound of variation of the gas price, and $\mu^{\mathrm{E}}-2T\sigma_{\mathrm{E}}+2(T-1)k\sigma_{\mathrm{E}}$ is the lower bound of variation of the electricity price. In (5) and (6), as long as the right hand side remain the same, the charging or discharging rate of storage would not change. Hence $\forall\epsilon>0$, given two policies π_t^1 and π_t^2 with property that: $E_t^1=E_t^2-\epsilon$, $A_t^{1,\mathrm{p2g}}=A_t^{2,\mathrm{p2g}}+\epsilon$ and $G_t^1=G_t^2+\frac{3.412}{\eta_{\mathrm{p2g}}}\epsilon$, the operation of storage is the same. This leads to that transition of these two policies, which will influence the expectation term, $T^1(S_t,\pi_t^1)$ and $T^2(S_t,\pi_t^2)$ is the same. The only difference of the revenue to go is the immediate revenue in period t.

$$\begin{aligned}
R(S_t,\pi_t^2) &= E_t^2\cdot P_t^{\mathrm{E}}+G_t^2\cdot P_t^{\mathrm{G}}\\
&=(E_t^1+\epsilon)\cdot P_t^{\mathrm{E}}+(G_t^1-\frac{3.412}{\eta_{p2g}}\epsilon)\cdot P_t^{\mathrm{G}}\\
&=R(S_t,\pi_t^1)+\epsilon\cdot(P_t^{\mathrm{E}}-\frac{3.412}{\eta_{p2g}}P_t^{\mathrm{G}})\\
&\geq R(S_t,\pi_t^1)+0\geq R(S_t,\pi_t^1).
\end{aligned}$$

This shows that if we generate more electricity by converting gas to electricity, the revenue to go would increase. Based on the assumption that the transmission line capacity is large enough, the optimal policy π_t^* should be where $A_t^{*\mathrm{p2g}}=-A_{\mathrm{g2p}}^{\max}$. A similar proof can be conducted to prove the second threshold. □

We say that the setting is an extreme case, when the highest price of one energy carrier is always lower than the lowest price of another energy carrier considering conversion efficiency. We can always make more money by converting more, which could be useful because the price of natural gas is usually about two or more times lower than the price of electricity for the same energy content in real life.

B. The Normal Case

Lemma 3. *For every period $t\in\mathbb{T}$, J_t is non-increasing in A_t^{p2g}, if $P_t^{\mathrm{E}}\geq 3.412/\eta_{\mathrm{p2g}}P_t^{\mathrm{G}}$, and the operation of storage keeps the same.*

Proof. In the proof of Theorem 2, we have shown that as long as the increase or decrease of E_t and G_t is determined by A_t^{p2g}, the state transition keeps the same. Given two such policies π_t^1 and π_t^2 with the property mentioned before, then $\mathbb{E}\left[J_{t+1}(S_{t+1})|(S_t,\pi_t^1)\right] = \mathbb{E}\left[J_{t+1}(S_{t+1})|(S_t,\pi_t^2)\right]$. If we assume $A_t^{\mathrm{p2g}} < 0$, the energy dynamic balance should hold during the energy conversion: $\frac{\partial E_t}{\partial A_t^{\mathrm{p2g}}} = -\frac{\eta_{\mathrm{p2g}}}{3.412}\frac{\partial G_t}{\partial A_t^{\mathrm{p2g}}}$. This relationship maps the increase of gas we buy to the increase of electricity we sell using energy conversion. The increase of the power of the convertor will increase the quantity of gas, which leads to $\frac{\partial G_t}{\partial A_t^{\mathrm{p2g}}} > 0$. By the chain rule, we can obtain the relation below:

$$\begin{aligned}\frac{\partial J_t}{\partial A_t^{\mathrm{p2g}}} &= \frac{\partial R_t}{\partial E_t}\frac{\partial E_t}{\partial A_t^{\mathrm{p2g}}} + \frac{\partial R_t}{\partial G_t}\frac{\partial G_t}{\partial A_t^{\mathrm{p2g}}} + \frac{\partial \mathbb{E}}{\partial A_t^{\mathrm{p2g}}}\\ &= P_t^{\mathrm{E}}\frac{\partial E_t}{\partial A_t^{\mathrm{p2g}}} + P_t^{\mathrm{G}}\frac{\partial G_t}{\partial A_t^{\mathrm{p2g}}}\\ &= (P_t^{\mathrm{G}} - \frac{\eta_{\mathrm{p2g}}}{3.412}P_t^{\mathrm{E}})\frac{\partial G_t}{\partial A_t^{\mathrm{p2g}}} \le 0,\end{aligned}$$

A similar proof can be conducted for $A_t^{\mathrm{p2g}} \ge 0$. Thus, J_t is non-increasing in A_t^{p2g}. □

Lemma 4. *For every period $t \in \mathbb{T}$, J_t is non-decreasing in A_t^{p2g}, if $P_t^{\mathrm{E}} \le 3.412\eta_{\mathrm{p2g}}P_t^{\mathrm{G}}$, and the operation of storage keeps the same.*

Proof. $\frac{\partial E_t}{\partial A_t^{\mathrm{p2g}}} = -\frac{\eta_{\mathrm{p2g}}}{3.412}\frac{\partial G_t}{\partial A_t^{\mathrm{p2g}}}$ or $\frac{\partial E_t}{\partial A_t^{\mathrm{p2g}}} = -\frac{1}{3.412\eta_{\mathrm{p2g}}}\frac{\partial G_t}{\partial A_t^{\mathrm{p2g}}}$, and since $0 < \eta_{\mathrm{p2g}} < 1$, $P_t^{\mathrm{G}} \ge \frac{1}{3.412\eta_{\mathrm{p2g}}}P_t^{\mathrm{E}} \ge \frac{1}{3.412}P_t^{\mathrm{E}} \ge \frac{\eta_{\mathrm{p2g}}}{3.412}P_t^{\mathrm{E}}$ holds $\forall t \in \mathbb{T}$.

$$\begin{aligned}\frac{\partial J_t}{\partial A_t^{\mathrm{p2g}}} &= \frac{\partial R_t}{\partial E_t}\frac{\partial E_t}{\partial A_t^{\mathrm{p2g}}} + \frac{\partial R_t}{\partial G_t}\frac{\partial G_t}{\partial A_t^{\mathrm{p2g}}} + \frac{\partial \mathbb{E}}{\partial A_t^{\mathrm{p2g}}}\\ &= P_t^{\mathrm{E}}\frac{\partial E_t}{\partial A_t^{\mathrm{p2g}}} + P_t^{\mathrm{G}}\frac{\partial G_t}{\partial A_t^{\mathrm{p2g}}}\\ &= (P_t^{\mathrm{G}} - \frac{\eta_{p2g}}{3.412}P_t^{\mathrm{E}})\frac{\partial G_t}{\partial A_t^{\mathrm{p2g}}} \ge 0,\end{aligned}$$

or

$$\frac{\partial J_t}{\partial A_t^{\mathrm{p2g}}} = (P_t^{\mathrm{G}} - \frac{1}{3.412\eta_{\mathrm{p2g}}}P_t^{\mathrm{E}})\frac{\partial G_t}{\partial A_t^{\mathrm{p2g}}} \ge 0.$$

□

Lemma 3 and 4 imply that we can obtain a higher revenue to go by decreasing or increasing the power of energy conversion, until some equality constraints become binding. Since we have assumed that the line capacities are large enough, A_t^{p2g} should reach its maximum or minimum.

Theorem 5. *There exists an optimal policy $\pi^* = (\pi_1^*, \ldots, \pi_T^*)$ characterized by two thresholds $\lambda_1 = 3.412\eta_{\mathrm{p2g}}$ and $\lambda_2 = \frac{3.412}{\eta_{\mathrm{p2g}}}$:*

1) *If $\frac{P_t^{\mathrm{E}}}{P_t^{\mathrm{G}}} \le \lambda_1$, then $A_t^{*\mathrm{p2g}} = -A_{\mathrm{g2p}}^{\max}$;*
2) *If $\frac{P_t^{\mathrm{E}}}{P_t^{\mathrm{G}}} \ge \lambda_2$, then $A_t^{*\mathrm{p2g}} = A_{\mathrm{p2g}}^{\max}$;*
3) *If $\lambda_1 < \frac{P_t^{\mathrm{E}}}{P_t^{\mathrm{G}}} < \lambda_2$, then $A_t^{*\mathrm{p2g}} = 0$.*

Proof. The first two cases are the results of Lemma 3 and 4. As for the last case, the following proof is given:

$$\frac{\partial J_t}{\partial A_t^{\mathrm{p2g}}} = \begin{cases}(P_t^{\mathrm{G}} - \frac{1}{3.412\eta_{\mathrm{p2g}}}P_t^{\mathrm{E}})\frac{\partial G_t}{\partial A_t^{\mathrm{p2g}}} < 0, & \text{if} A_t^{\mathrm{p2g}} \ge 0,\\ (P_t^{\mathrm{G}} - \frac{\eta_{\mathrm{p2g}}}{3.412}P_t^{\mathrm{E}})\frac{\partial G_t}{\partial A_t^{\mathrm{p2g}}} > 0, & \text{if} A_t^{\mathrm{p2g}} < 0.\end{cases}$$

Hence the maximum happens at where $A_t^{*\mathrm{p2g}} = 0$. □

Theorem 5 reduces one decision variable $A_t^{*\mathrm{p2g}}$ that needs to be determined, according to the realization of S_t. For every S_t and every t, by Theorem 5, the decision variables left are the optimal operation of electric and gas storage.

Lemma 6. *For every period $t \in \mathbb{T}$, J_t is non-decreasing in IE_t and IG_t, given any $\mathbf{P_t} = (P_t^{\mathrm{G}}, P_t^{\mathrm{E}})$ and W_t.*

Proof. We prove J_t is non-decreasing in IE_t first. When $t = T$, since the boundary condition we defined is $J_{T+1}(S_{T+1}) = 0, \forall S_{T+1}$. The expectation term at $t = T$ is zero, which leads to: $J_T(S_T) = R(S_T, \pi_T)$. For $\pi_T^* = \mathrm{argmax}_{\pi\in\Pi} R(S_T, \pi_T)$, $R_T^* = E_T^* P_T^{\mathrm{E}} + G_T^* P_T^{\mathrm{G}}$ holds. Using (5) and (6), we can conclude that $R_T^* = W_T P_T^{\mathrm{E}} - A_T^{*\mathrm{p2g}} P_T^{\mathrm{E}} - A_T^{*\mathrm{e}} P_T^{\mathrm{E}} + F_T^{*\mathrm{p2g}} P_T^{\mathrm{G}} - A_T^{*\mathrm{g}} P_T^{\mathrm{G}}$. The storage should be on discharging mode at T. For A_T^{e}, we have $|A_T^{\mathrm{e}}| \le \min(A_{\mathrm{e}}^{\max}, IE_T)$, and $|A_T^{*\mathrm{e}}| = \min(A_{\mathrm{e}}^{\max}, IE_T)$. Assume $\pi_T^* = \mathrm{argmax}_{\pi\in\Pi} R(S_T, \pi_T)$ and $\pi_T^{*'} = \mathrm{argmax}_{\pi\in\Pi} R(S_T', \pi_T)$, where the only difference of S_T and S_T' is IE_T and IE_T', $|A_T^{*\mathrm{e}}| \le |A_T^{*\mathrm{e}'}|$ holds. $R_T^* = W_T P_T^{\mathrm{E}} - A_T^{*p2g} P_T^{\mathrm{E}} - A_T^{*\mathrm{e}} P_T^{\mathrm{E}} + F_T^{*\mathrm{p2g}} P_T^{\mathrm{G}} - A_T^{*\mathrm{g}} P_T^{\mathrm{G}} \le W_T P_T^{\mathrm{E}} - A_T^{*\mathrm{p2g}} P_T^{\mathrm{E}} - A_T^{*\mathrm{e}'} P_T^{\mathrm{E}} + F_T^{*\mathrm{p2g}} P_T^{\mathrm{G}} - A_T^{*\mathrm{g}} P_T^{\mathrm{G}} = R_T^{*'}$, i.e. in period T, the hypothesis holds.

Let us assume that the hypothesis holds for $T, T-1, \ldots, t+1$, $\pi_t^* = \mathrm{argmax}_{\pi_t^*\in\Pi} R(S_t, \pi_t) + \mathbb{E}\left[J_{t+1}(S_{t+1})|S_t\right]$, and consider a small variation $\epsilon > 0$ in IE_t, i.e., $IE_t' = IE_t + \epsilon$, if the storage is in discharging mode, there exits a new policy π_t with $A_t^{\mathrm{e}} \in [A_t^{*\mathrm{e}} - \epsilon, A_t^{*\mathrm{e}}]$ that $IE_{t+1}(S_t', \pi_t) = LE_{t+1}(S_t', \pi_t) = \max(0, IE_t' + A_t^e) \ge \max(0, IE_t + A_t^{*e}) = IE_{t+1}(S_t, \pi_t^*)$. For the two different policies with different states, $A_t^{\mathrm{e}} \le A_t^{*\mathrm{e}} \le 0$ holds, which leads to $R(S_t', \pi_t) \ge R(S_t, \pi_t^*)$. We can conclude that if there is a variation, there exits a new policy that:

$$\begin{aligned}J_t(S_t', \pi_t) &= R(S_t', \pi_t) + \mathbb{E}\left[J_{t+1}(S_{t+1}')|S_t'\right]\\ &\ge R(S_t', \pi_t) + \mathbb{E}\left[J_{t+1}(S_{t+1})|(S_t, \pi_t^*)\right]\\ &\ge R(S_t, \pi_t^*) + \mathbb{E}\left[J_{t+1}(S_{t+1})|(S_t, \pi_t^*)\right]\\ &= J_t(S_t, \pi_t^*).\end{aligned}$$

For variation $\epsilon > 0$ in IE_t, we can always find a new policy that $J_t(S_t', \pi_t) \ge J_t(S_t, \pi_t^*)$. A similar process can be applied for the charging mode. Let $\pi_t^{*'} = \mathrm{argmax}_{\pi_t^{*'}\in\Pi} R(S_t', \pi_t) + \mathbb{E}\left[J_{t+1}(S_{t+1})|S_t'\right]$, $J_t(S_t', \pi_t^{*'}) \ge J_t(S_t', \pi_t) \ge J_t(S_t, \pi_t^*)$, the hypothesis holds for t. J_t is non-decreasing in IE_t. A simular proof can be conducted to prove J_t is non-decreasing in IG_t. □

For convenience, we change R_t into the following equation using equality constraint:

$$R_t^* = W_t P_t^{\mathrm{E}} - A_t^{*\mathrm{p2g}} P_t^{\mathrm{E}} - A_t^{*\mathrm{e}} P_t^{\mathrm{E}} + F_t^{*\mathrm{p2g}} P_T^{\mathrm{G}} - A_t^{*\mathrm{g}} P_t^{\mathrm{G}}. \tag{14}$$

The W_t can be regarded as known after the state realization, and Theorem 5 reduces one decision variable. The problem can be simplified into a new problem:

$$V_t(S_t) = \max_{A_{\mathrm{e},t} \in \mathbb{A}_{\mathrm{e}}, A_{\mathrm{g},t} \in \mathbb{A}_{\mathrm{g}}} \left\{ -A_t^{\mathrm{e}} P_t^{\mathrm{E}} - A_t^{\mathrm{g}} P_t^{\mathrm{G}} + \mathbb{E}[V_{t+1}|S_t] \right\}. \tag{15}$$

Let us define the positive and negative parts of A_t^{e} and A_t^{g} as:

$$\begin{aligned} &e_t^+ = \max(0, A_t^{\mathrm{e}}),\ e_t^- = -\min(0, A_t^{\mathrm{e}}); \\ &g_t^+ = \max(0, A_t^{\mathrm{g}}),\ g_t^- = -\min(0, A_t^{\mathrm{g}}); \\ &V_t(S_t) = \max\{r(a_t, S_t) + \mathbb{E}[V_{t+1}|S_t]\}, \end{aligned}$$

where $r(a_t, S_t) = -e_t^+ P_t^{\mathrm{E}} - g_t^+ P_t^{\mathrm{G}} + e_t^- P_t^{\mathrm{E}} + g_t^- P_t^{\mathrm{G}}$; a_t is the collection of e_t^+, e_t^-, g_t^+ and g_t^-.

Lemma 7. *For every period $t \in \mathbb{T}$, V_t is concave in IE_t and IG_t, given any $\mathbf{P_t} = (P_t^{\mathrm{G}}, P_t^{\mathrm{E}})$ and W_t.*

Proof. We would use backward induction to prove IE_t first. For the terminal stage T, the expectation term is zero, and the discharging rate is limited by the rate capacity and IE_T, i.e., $e_T^- = \max(A_e^{\max}, IE_T)$. Since the expectation term is zero and the rest is affine, V_T is concave.

Assume V_t is concave for $t = T, T-1, \ldots, t+1$, we need to prove $\frac{V_t^*(S_t^1)+V_t^*(S_t^2)}{2} \leq V_t^*(S_t^3)$, where $2 \cdot IE_t^3|S_t^3 = IE_t^1|S_t^1 + IE_t^2|S_t^2$. Note that the feasible sets of our decision variables are convex sets because all constraints are affine, thus any convex combination of any two feasible decisions should be feasible too. Let $\pi_t^{*1} = \operatorname{argmax} V_t(S_t^1)$ and $\pi_t^{*2} = \operatorname{argmax} V_t(S_t^2)$, then $\pi_t^3 = \frac{\pi_t^{*1}+\pi_t^{*2}}{2}$ is feasible. All states excluding IE_{t+1} will not change associated with these three actions. Let S_{t+1}^3, S_{t+1}^2 and S_{t+1}^1 be the states of the next stage by applying π_t^3, π_t^{*1} and π_t^{*2} respectively. We have $S_{t+1}^3 = \frac{S_{t+1}^2+S_{t+1}^1}{2}$, and by concavity in period $t+1$:

$$\begin{aligned} &\mathbb{E}[V_{t+1}(S_{t+1}^1)] + \mathbb{E}[V_{t+1}(S_{t+1}^2)] \leq 2\mathbb{E}[V_{t+1}(S_{t+1}^3)]; \\ &r(\pi_t^{*1}, S_t^1) + r(\pi_t^{*2}, S_t^2) = 2r(\pi_t^3, S_t^3); \\ &\frac{V_t^*(S_t^1, \pi_t^{*1}) + V_t^*(S_t^2, \pi_t^{*2})}{2} \leq V_t(S_t^3, \pi_t^3) \leq V_t^*(S_t^3, \pi_t^{*3}). \end{aligned}$$

$V_t(S_t)$ is concave in IE_t in period t, which means the lemma holds for t. In conclusion, V_t is concave in IE_t. A similar proof can be conducted to prove that V_t is concave in IG_t. □

Lemma 7 implies that there exists a stationary policy for the revenue to go function in the Bellman equation. Using this, we observe that the optimal policy has a dual-threshold structure. In order to check the first order condition of V_t, we define $\bar{V}_{t+1}(S_{t+1}) = \mathbb{E}[V_{t+1}(S_{t+1})|(S_t, \pi_t)]$, which leads to the equation below:

$$\begin{aligned} V_t(S_t) = \max\{ &-e_t^+ P_t^{\mathrm{E}} - g_t^+ P_t^{\mathrm{G}} + e_t^- P_t^{\mathrm{E}} \\ &+ g_t^- P_t^{\mathrm{G}} + \bar{V}_{t+1}(S_{t+1})\}. \end{aligned} \tag{16}$$

Consider the optimality of concave function, the following first order necessary and sufficient condition should hold: $\frac{\partial V_t(S_t)}{\partial S_t} = 0$. We focus on IE_t and IG_t, let us rewrite the condition as $\frac{\partial V_t(IE_t)}{\partial IE_t} = 0$ and $\frac{\partial V_t(IG_t)}{\partial IG_t} = 0$. Since V_t is concave in IE_t and in IG_t, $\bar{V}_{t+1}(S_{t+1})$ is concave, we can have its right and left directional derivatives on IE_t and IG_t. Let $\partial_{IE}^+ \bar{V}_t(S_t)$ and $\partial_{IE}^- \bar{V}_t(S_t)$ to be two directional derivatives on IE and $\partial_{IG}^+ \bar{V}_t(S_t)$, and $\partial_{IG}^- \bar{V}_t(S_t)$ to be two directional derivatives on IG.

Lemma 8. *Given the current system state S_t, there is no electric or gas storage operation, i.e. $A_t^{\mathrm{e}} = 0$ or $A_t^{\mathrm{g}} = 0$, if the following conditions hold simultaneously:*

$$\begin{cases} -P_t^{\mathrm{E}} + \eta_{\mathrm{e}} \partial_{IE}^+ \bar{V}_t(S_{t+1}) & \leq 0, \\ P_t^{\mathrm{E}} + \frac{1}{\eta_{\mathrm{e}}} \partial_{IE}^- \bar{V}_t(S_{t+1}) & \leq 0; \end{cases}$$

or

$$\begin{cases} -P_t^{\mathrm{G}} + \eta_{\mathrm{g}} \partial_{IG}^+ \bar{V}_t(S_{t+1}) & \leq 0, \\ P_t^{\mathrm{G}} + \frac{1}{\eta_{\mathrm{g}}} \partial_{IG}^- \bar{V}_t(S_{t+1}) & \leq 0. \end{cases}$$

Proof. We write the first-order conditions $\frac{\partial V_t(IE_t)}{\partial IE_t} = 0$ in terms of the action e_t^+ and e_t^- to maximize the right hand side of the Bellman equation:

$$\begin{aligned} -P_t^{\mathrm{E}} + \eta_e \partial_{IE}^+ \bar{V}_t(S_{t+1}) &\leq 0, && \text{if, } e_t^+ \geq 0, \\ -P_t^{\mathrm{E}} + \eta_e \partial_{IE}^- \bar{V}_t(S_{t+1}) &\geq 0, && \text{if, } e_t^+ > 0, \\ P_t^{\mathrm{E}} + \frac{1}{\eta_e} \partial_{IE}^- \bar{V}_t(S_{t+1}) &\leq 0, && \text{if, } e_t^- \geq 0, \\ P_t^{\mathrm{E}} + \frac{1}{\eta_e} \partial_{IE}^+ \bar{V}_t(S_{t+1}) &\geq 0, && \text{if, } e_t^- > 0. \end{aligned} \tag{17}$$

Because we defined e_t^+ and e_t^- as $e_t^+ = \max(0, A_t^{\mathrm{e}})$, $e_t^- = -\min(0, A_t^{\mathrm{e}})$, $A_t^{\mathrm{e}} = 0$ holds, if and only if $e_t^+ = 0$ and $e_t^- = 0$ hold simultaneously. Hence if $A_t^{\mathrm{e}} = 0$ holds:

$$-P_t^{\mathrm{E}} + \eta_e \partial_{IE}^+ \bar{V}_t(S_{t+1}) \leq 0, \tag{18}$$

$$P_t^{\mathrm{E}} + \frac{1}{\eta_e} \partial_{IE}^- \bar{V}_t(S_{t+1}) \leq 0 \tag{19}$$

□

Lemma 8 shows that there exits two thresholds $k_t(S_t)$ and $h_t(S_t)$ that characterize an optimal policy. Let $k_t(S_t)$ be the maximum IE_t or IG_t that (19) holds, if (19) does not hold even for $IE_t = 0$ or $IG_t = 0$, then $k_t(S_t) = 0$; Similarly, let $h_t(S_t)$ be the minimum IE_t or IG_t that (18) holds, if (18) does not hold even for $IE_t = IE_{\max}$ or $IG_t = IG_{\max}$, then $k_t(S_t) = IE_{\max}$ or $k_t(S_t) = IG_{\max}$.

Theorem 9. *There exists an optimal policy $\pi^* = (\pi_1^*, \ldots, \pi_T^*)$ characterized by two thresholds $k_t^{\mathrm{e}}(S_t)$ and $h_t^{\mathrm{e}}(S_t)$ or $k_t^{\mathrm{g}}(S_t)$ and $h_t^{\mathrm{g}}(S_t)$.*

1) *If $IE_t \leq h_t^{\mathrm{e}}(S_t)$ or $IG_t \leq h_t^{\mathrm{g}}(S_t)$, then greedily charge the storage to $h_t^{\mathrm{e}}(S_t)$ or $h_t^{\mathrm{g}}(S_t)$, and $A_t^{*\mathrm{e}} = \min(\frac{A_{\mathrm{e}}^{\max}}{\eta_{\mathrm{e}}}, \frac{h_t^{\mathrm{e}}(S_t) - IE_t}{\eta_{\mathrm{e}}})$, $A_t^{*\mathrm{g}} = \min(\frac{A_{\mathrm{g}}^{\max}}{\eta_{\mathrm{g}}}, \frac{h_t^{\mathrm{g}}(S_t) - IE_t}{\eta_{\mathrm{g}}})$.*

2) *If* $IE_t \geq k_t^{\mathrm{e}}(S_t)$ *or* $IG_t \geq k_t^{\mathrm{g}}(S_t)$*, then greedily discharge the storage to* $k_t^{\mathrm{e}}(S_t)$ *or* $k_t^{\mathrm{g}}(S_t)$ *and* $A_t^{*\mathrm{e}} = \max(\eta_{\mathrm{e}} A_{\mathrm{e}}^{\max}, \eta_{\mathrm{e}}(IE_t - k_t^{\mathrm{e}}(S_t)))$*,* $A_t^{*\mathrm{g}} = \max(\eta_{\mathrm{g}} A_{\mathrm{g}}^{\max}, \eta_{\mathrm{g}}(IG_t - k_t^{\mathrm{g}}(S_t)))$*.*

Proof. Let $IE_t^{'} = h_t^{\mathrm{e}}(S_t)$, $\forall IE_t \leq IE_t^{'}$, by concavity we have $\partial_{IE}^{+}\bar{V}_t(IE_t^{'}) \leq |\partial_{IE}^{-}\bar{V}_t(IE_t)|$, which leads to $-P_t^{\mathrm{E}} + \eta_{\mathrm{e}}\partial_{IE}^{-}\bar{V}_t(S_{t+1}) \geq 0$. Thus, $e_t^{+} > 0$; $\forall IE_t \geq k_t^{\mathrm{e}}(S_t)$, we have $P_t^{\mathrm{E}} + \frac{1}{\eta_e}\partial_{IE}^{+}\bar{V}_t(S_{t+1}) \geq 0$. Thus, $e_t^{-} > 0$. A similar proof can be conducted for the gas storage. □

The thresholds above rely on the computation of the partial derivatives. Following the same approach of Theorem 4.1 in [10], it can be shown that the directional derivatives are piecewise linear in IE_t or IG_t, $\forall t \in \mathbb{T}$.

To capture the impact of the power conversion, the following modification is needed for thresholds:

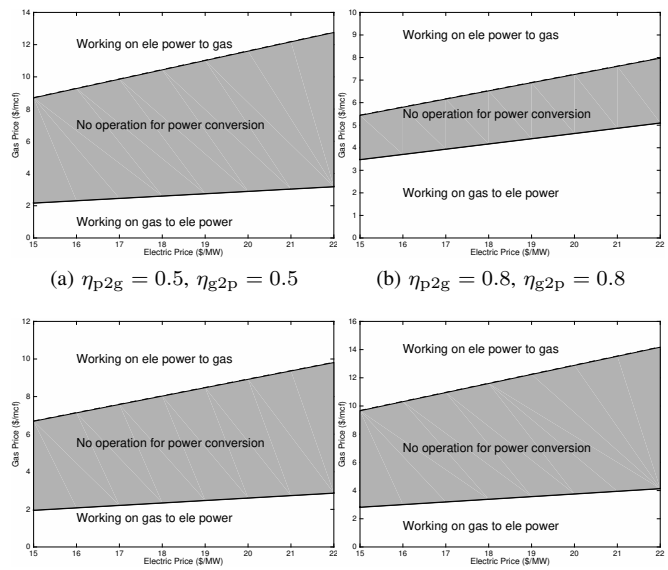
$$P_t^{\mathrm{E}} = \min(P_t^{\mathrm{E}}, 3.412\eta_{\mathrm{p2g}} P_t^{\mathrm{G}}),\ P_t^{\mathrm{G}} = \min(P_t^{\mathrm{G}}, \frac{\eta_{\mathrm{p2g}}}{3.412} P_t^{\mathrm{E}}).$$

IV. Numerical Examples

We conduct some examples with different efficiencies to analyze the feasible region of the operation of the energy conversion using data from [9]. Let η_{p2g} be the efficiency of the energy conversion if it works on electric power to gas mode; η_{g2p} be the efficiency of the energy conversion if it works on gas to electric power mode. As shown in Fig. 2, when the efficiencies of P2G and G2P are high, the region of no operation is small. Because the efficiency means energy loss, consider the case where the efficiency is 1, there is no energy loss during energy conversion, and then the threshold would be the same line. For that case, energy conversion will bring more economic value unless the price is the same for the same energy content. The larger efficiency of G2P leads to a larger threshold of the gas to electric power conversion. When the efficiency is low, which means the energy loss is high, a comparably lower gas price is required if we want to get benefits by buying gas, converting gas to electricity, and selling electricity. Similarly when the efficiency of P2G is low, a comparably higher gas price is required if we want to get benefits by buying electricity, converting electricity to gas, and selling gas. Hence the larger efficiency leads to a lower threshold of the electric power to gas conversion.

V. Conclusion

We study the optimal operation of a profit-maximizing energy hub equipped with the energy conversion and the energy storage through a dynamic programming formulation. For an extreme setting that incorporates uncertainty, we prove the optimality of a two-threshold policy of the energy conversion. Furthermore, we propose the two-threshold policies of the energy conversion and the energy storage respectively in normal cases. For a setting with parameters that leads to the normal cases, we show that the optimal operation of the energy conversion depends on the efficiency of the conversion process. We also show that the expected net benefit is a concave function of the state of charge of storage, which leads to a stationary policy for the operation of the energy storage. To capture the impact of efficiency and energy conversion, we use two stationary thresholds rather than one stationary threshold for the storage, and modify the prices accordingly.



(a) $\eta_{\mathrm{p2g}} = 0.5$, $\eta_{\mathrm{g2p}} = 0.5$ (b) $\eta_{\mathrm{p2g}} = 0.8$, $\eta_{\mathrm{g2p}} = 0.8$

(c) $\eta_{\mathrm{p2g}} = 0.65$, $\eta_{\mathrm{g2p}} = 0.45$ (d) $\eta_{\mathrm{p2g}} = 0.45$, $\eta_{\mathrm{g2p}} = 0.65$

Fig. 2: The feasible region of each operation in the normal cases.

In future work, we will investigate the efficiency in detail where the efficiency is a function of decisions, and we will explore nonlinear pricing schemes by extending our dynamic programming framework.

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